

FU Xiao

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EDUCATION

The Hong Kong University of Science and Technology Clear Water Bay, Kowloon, Hong Kong
Doctor of Philosophy(Ph.D.) in Computer Science and Engineering Sept. 2022 – Jul. 2026 (Incoming)

- **Supervisor:** Prof. [Dan Xu](#)
- **Research Interests:** (1) 3D Computer Vision: Self-supervised Scene Geometry Learning, joint Learning of Implicit and Explicit Scene Representation, (2)Machine Learning

Zhejiang University Hangzhou, Zhejiang
Bachelor in Information Engineering | College of Information Science & Electronic Engineering Sept. 2018 – Jul. 2022

- **GPA:** 3.96/4.00, 90.45/100 (92.33/100 for 2nd year), **Ranking:** 6/146
- **Chu Kochen Honors College:** Intensive Training Program of Innovation and Entrepreneurship (ITP) (Honored minor for the selected top 40 students with innovation capability from 6,000 freshmen)

PUBLICATIONS

* indicates equal contribution

- [C1] Guangyi Zhang*, **Xiao Fu***, Qiyu Hu, Yunlong Cai, Guanding Yu. Hybrid Precoding Design for Multiuser MIMO Systems Based on Dual-Layer Deep-Unfolding Neural Network. *IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2021 [[Paper](#)]
(Two strong acceptances out of three random reviewers. Held in Helsinki, Finland)

RESEARCH EXPERIENCE

Dynamic Perceptive Network for Monocular Depth Estimation Jul. 2021 – Sept. 2021
Supervisor: Prof. [Xiaojuan Qi](#) CVMI Lab, The University of Hong Kong

- Propose a neural network based on dynamic routing framework, which could generate adaptive receptive field size in response to arbitrary resolution of input images, tackling the problem of the inconsistent structure in high-resolution monocular depth estimation
- For one high-resolution(1443×1082 for IBMS-1) depth estimation on GPU Tesla V10, OURS consumes 1.03s compared to 11.24s for SoTA, which means our model requires less computational complexity
- As a tradeoff for less complexity and faster deducement, the performance of OURS drops around 4% on ordinal error(ORD) and RMSE metrics, tested on datasets:Middlebury 2014 and IBMS-1

Machine Learning for Wireless Network Complexity Optimization Apr. 2020 – Dec. 2020
Supervisor: Prof. [Guanding Yu](#) Zhejiang University

- Proposed a novel framework, i.e. a Dual-Layer Deep-Unfolding Neural Network(DLDUNN) which unfolds the iterative Penalty Dual Decomposition (PDD) algorithm into a layer-wise structure whereby some trainable parameters are introduced into the places of the high-complexity operations, e.g. large-dimensional matrix inversions and iterative sub algorithms, so as to address the high computational complexity problem of the dual-layer PDD algorithm and presents the major original performance in the meantime
- **Laureate of Excellent Award(Team leader) - Student Research Training Program(Province-level)**

SELECTED HONORS AND SCHOLARSHIPS [[Certificate & Report](#)]

- National Scholarship (**Highest Honor for Chinese Undergraduates, Top 1.8%**) 2020
- ISEE Yuelun Alumni Scholarship (**6/1,150 selection ratio, Bonus 20,000 CNY**) 2020
- First-class Scholarship for Academic Excellence of Zhejiang University (**Top 3%**) 2020
- The First Prize of National Talents Training Base (**Top 3%**) 2020
- Government Scholarship for Academic Excellence of Zhejiang Province (**Top 5%**) 2019
- Finalist Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 2% out of 20,954 teams**) 2020
- Meritorious Winner – Mathematical Contest in Modeling (MCM/ICM) (**Top 7% out of 26,112 teams**) 2021
- Gold Prize – 7th Zhejiang International “Internet+” Innovation and Entrepreneurship Contest (**1st author**) 2021
- Gold Prize – 12th “Challenge Cup” National College Student Business Plan Competition (**4th author**) 2021
- Grand Prize – “Yongdian Cup” Innovation and Entrepreneurship Contest (**1st author**) 2019
- Second Prize – National College Mathematics Competition 2019
- Second Prize – Zhejiang Province Physics Theory (Innovation) Competition 2019
- Zhejiang University Five-Star Volunteer 2021

TECHNICAL SKILLS

Programming: Pytorch/Tensorflow, Python, C/C++, LaTeX, MATLAB, Verilog

Developer Tools: Git, VS Code, Visual Studio, PyCharm

Motto: Don't regret anything in life. If it's good, it's wonderful. If it's bad, it's experience! When you win, you win.
When you lose, you learn. — *Prof. Feynman*